

V01.04 – Mirror Mathematics I

A Constraint Calculus for Recursive Observerhood

Formal Preliminaries for the Mathematics Arc of the Mirror Programme

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Mirror Programme, Volume I: Observerhood – Mathematics Arc

Abstract

Mirror Theory I–III developed the conceptual theory of derived observerhood: observerhood is not treated as a primitive subject, point of view or metaphysical witness, but as a regime of bounded systems that maintain world-models, self-models and reliability estimates under viability constraints. The present paper supplies the first mathematical layer of that programme. It defines a constraint regime $C = (S, T, \mathcal{I}, V, \kappa)$, consisting of a state space, an admissible transition structure, a family of invariants, a viability or persistence functional and a bounded capacity parameter.

The central purpose of the calculus is to make the programme’s formal grammar explicit. A system is not an observer merely because it persists, computes, predicts, regulates or contains information. The Mirror predicate requires bounded local modelling, viability-relevant self-modelling and a positive salience gain for reliability information after cost.

This July 2026 edition aligns the paper with the Volume I roadmap: the Theory papers supply the conceptual derivation, the Mathematics papers formalise the threshold structure, the Observerhood Labs provide computational scaffolding, and the Physics papers analyse physical support conditions. The main results remain deliberately narrow. The Necessary Constraint Theorem identifies transition structure, invariant preservation, bounded compression and viability relevance as necessary but not sufficient conditions for deriving a stable observerhood predicate. The Constraint-Salience Threshold Theorem formalises the point at which self-model reliability becomes observer-relevant. The Observer-Equivalence Proposition explains when differently implemented systems count as the same observer-type at the level of maintained constraint organisation. The paper does not modify physics, solve consciousness or replace existing mathematical frameworks; it supplies the formal vocabulary in which such later questions can be asked responsibly.

Keywords: constraint regimes; observerhood; viability; recursive reliability; bounded modelling; salience; active inference; cybernetics; information theory; Mirror Theory.

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1 Roadmap position

The Mirror Programme is organised as a nested research architecture: programme, cycles, volumes and papers. Its programme-level spine is constraint, model, reliability, repair, memory, viability, continuity and scale [1]. Volume I applies that spine to observerhood and treats observerhood as a derived organisational condition rather than a primitive witness, subject or metaphysical point of view [2].

Within Volume I, the Theory arc has already supplied the conceptual foundation. Mirror Theory I gives the minimal computational derivation; Mirror Theory II develops philosophical consequences for identity, meaning, persistence and reality; and Mirror Theory III situates the framework among neighbouring accounts in cognitive science, philosophy of mind and artificial intelligence [3, 4, 5]. The present paper begins the Mathematics arc. Its task is not to repeat the theory papers, but to provide the formal grammar needed by the later threshold, memory-governance, laboratory and physical-support papers.

2 Core claim

The core claim of this paper is deliberately limited:

A bounded local subsystem crosses the minimal Mirror observer threshold only when it maintains a world-model, maintains a self-model, represents or uses reliability information about that self-model, and obtains positive expected viability gain from that reliability information after cost.

The claim is structural, not phenomenological. It does not prove consciousness. It does not assign observerhood to all information-processing systems. It does not propose a new physical law. It states the formal dependency structure that must be satisfied before recursive observerhood can be treated as a disciplined predicate within the Mirror Programme.

3 Introduction

Mirror Theory began as a theory of observerhood rather than a theory of consciousness. Its methodological rule was simple: an observer may not be smuggled into the premises. If observerhood is to be used as an explanatory category, it must be derived from more primitive structure.

The earlier Theory papers developed that programme in stages. Mirror Theory I gave the local computational derivation. Mirror Theory II explored identity, meaning, death and realism under derived observerhood. Mirror Theory III compared the framework with neighbouring theories and clarified what Mirror Theory does and does not claim [3, 4, 5]. The present paper now asks a preparatory mathematical question: if observerhood is a threshold in constraint organisation, what formal objects are required before the programme can responsibly move into experiments, memory governance and physical support?

This paper answers that question by defining a calculus of constraint regimes: state spaces equipped with admissible transitions, invariants, viability functionals and capacity bounds. The aim is to express, with as little metaphysical language as possible, the formal grammar behind the Mirror Programme. This paper sharpens three elements: concrete operational readings of the viability functional V , a closer mapping to active inference, predictive processing and cybernetics, and a minimal hierarchy of reliability variables suitable for artificial-agent experiments.

The core intuition is this:

A world capable of supporting observers must be stable enough for persistence, structured enough for compression, regular enough for prediction, bounded enough to require modelling, and consequential enough that errors in self-location can matter.

This sentence is not yet physics. It is a constraint on any future physical extension. If a proposed universe contains no invariant-preserving transitions, no stable substructures, no local capacity bounds and no viability-relevant distinctions, then observerhood cannot be derived inside it. If a bounded subsystem models a world, models itself inside that world, and gains viability by modelling the reliability of its self-location, then the Mirror observerhood predicate becomes licensed.

This paper therefore formalises the layer between conceptual theory and any future physical-support work. It is the mathematical scaffolding that must exist before one writes responsibly about causal speed, spacetime, quantum information or physical constraint salience within the Mirror Programme.

4 Scope and negative commitments

The paper’s scope is narrow. It does not claim to solve physics, consciousness or the origin of reality. It defines objects that may later be useful in those discussions.

4.1 Not a modification of relativity

No claim is made here that the invariant speed c varies, that faster-than-light travel is possible, or that spacetime geometry must be rewritten. Any future Mirror-compatible physics would have to reduce to ordinary relativity in all tested domains and generate at least one new empirical prediction. This paper does neither. It supplies preliminary mathematical language.

4.2 Not a theory of consciousness

The calculus concerns observerhood as a structural predicate. It does not prove phenomenality, subjective feeling or consciousness in the hard-problem sense. A system may satisfy formal observerhood criteria while leaving phenomenal consciousness as an additional question.

4.3 Not panpsychism

The calculus does not assign observerhood to every persistent structure. Rocks, crystals, thermostats, feedback loops and self-referential strings may all be describable in constraint-theoretic terms without satisfying the observer threshold.

4.4 Not a replacement for existing mathematics

The framework borrows from dynamical systems, information theory, control theory and category-theoretic thinking [8, 7, 6, 9]. It does not replace those frameworks. It repackages the specific pieces needed for Mirror Theory: persistence, invariance, compression, viability and recursive self-location.

5 Constraint regimes

Definition 5.1 (Constraint regime). *A constraint regime is a tuple*

$$C = (S, T, \mathcal{I}, V, \kappa),$$

where:

- (i) S is a state space;
- (ii) $T \subseteq S \times S$ is an admissible transition relation, or more generally a transition operator on states or probability measures over states;
- (iii) $\mathcal{I}$ is a family of invariants, functions, relations or structures preserved under some class of admissible transitions;
- (iv) $V : S^{\mathbb{N}} \times H \rightarrow \mathbb{R}$ is a viability or persistence functional assigning value to trajectories over a horizon H according to maintenance of a relevant organisation;
- (v) κ is a capacity bound on encoding, modelling or processing.

The regime C is not assumed to be physical. It may describe a physical system, an artificial agent, a biological organism, a symbolic system or an abstract dynamical structure. The object is deliberately general.

Definition 5.2 (Admissible trajectory). *A sequence $\tau = (s_0, s_1, s_2, \dots) \in S^{\mathbb{N}}$ is an admissible trajectory of C if $(s_t, s_{t+1}) \in T$ for all t , or if it is generated by the transition operator T where T is stochastic or measure-valued.*

Definition 5.3 (Invariant). *An invariant $J \in \mathcal{I}$ is a function, relation or structure such that for an admissible transition $(s, s') \in T$, preservation holds in the relevant sense:*

$$J(s) = J(s'),$$

or, more generally, J is preserved up to a specified equivalence relation or tolerance.

The tolerance clause matters. Biological, cognitive and social systems rarely preserve exact numerical identity. They preserve organisation. A living body changes matter while preserving metabolic structure. A theory changes notation while preserving inferential content. An observer changes atoms while preserving the recursive organisation that maintains its identity.

Definition 5.4 (Stable substructure). *A subset $A \subseteq S$ is a stable substructure relative to T and $\mathcal{I}$ if admissible transitions beginning in A remain in, return to, or preserve an equivalence class of A under the relevant invariants for a non-trivial interval of transition order.*

Remark 5.1. *This definition covers fixed points, attractors, limit cycles, metastable regimes and identity-preserving transformations. It is intentionally broad because Mirror Theory concerns persistence of organisation rather than only equilibrium.*

6 Operational readings of the viability functional

The viability functional V does substantial work in the calculus. In the abstract it denotes organisational preservation, but empirical work requires more concrete readings. The point of V is not to introduce a hidden purpose. It identifies the quantity, probability or ordering with respect to which some organisational pattern persists, degrades, repairs or fails.

Definition 6.1 (Horizon-relative viability). *For a local subsystem L , let H be a prediction or persistence horizon. A horizon-relative viability functional is a map*

$$V_L : S_L^H \rightarrow \mathbb{R}$$

assigning value to possible trajectories of L over H according to the continued maintenance of the relevant organisation.

In artificial agents, V_L may be implemented as expected discounted reward, task-completion probability, battery preservation, mission survival, constraint-satisfaction score, or probability of remaining within an operational envelope. In biology, it may track metabolic stability, cellular integrity, homeostatic regulation, reproductive persistence or survival probability. In distributed systems, it may track uptime, redundancy, connectivity, error tolerance or continuation of a protocol. In cognitive systems, it may track preservation of agency, memory integrity, planning capacity, bodily regulation or coherent self-location.

A useful generic form is

$$V_L(\tau, H) = \mathbb{E}[\text{Pr}(\text{organisation of type } L \text{ remains viable through } H) \mid \tau] - C(\tau),$$

where $C(\tau)$ is a cost term for energy, computation, risk, degradation or opportunity loss. In reinforcement-learning settings this may reduce to a reward functional. In biological settings it may reduce to a survival or homeostatic functional. In symbolic systems it may reduce to preservation of inferential coherence or goal-satisfaction under perturbation.

This mapping matters because the salience threshold $\Sigma_L(R_L) > 0$ is only meaningful once V_L is specified. Without a measurable viability functional, the calculus remains philosophical notation. With one, the threshold becomes experimentally addressable: remove or perturb a reliability variable and ask whether viability over horizon H changes.

Domain	Possible V_L	Example measurement
Artificial agent	Long-horizon reward, mission persistence, operational envelope	Task success after sensor drift; battery remaining; recovery after memory corruption
Biological organism	Homeostasis, survival probability, metabolic integrity	Regulation after injury, infection, hunger, fatigue or disorientation
Cellular system	Membrane integrity, metabolic persistence, reproduction	ATP balance, membrane maintenance, replication rate
Distributed system	Network survival, redundancy, error tolerance	Uptime, connectivity, fault recovery, consistency after node failure
Scientific community	Model correction and persistence under evidence	Replication success, error detection, instrument calibration, theory revision

Table 1: Operational readings of the viability functional. The formal symbol V_L is deliberately abstract, but each experimental domain must instantiate it with measurable quantities.

7 Local subsystems and compression

Observerhood is never assigned to a whole state space merely because it has structure. It is assigned, if at all, to bounded local subsystems inside a regime.

Definition 7.1 (Local subsystem). *Let $C = (S, T, \mathcal{I}, V, \kappa)$. A local subsystem L is a distinguished factor, subset, subspace or coarse-graining S_L of S , dynamically coupled to an environment E such that S may be represented, exactly or approximately, as $S_L \times S_E$ or by a coupling relation between S_L and S_E .*

Definition 7.2 (Capacity bound). *A subsystem L is capacity-bounded if the number or complexity of states relevant to its future trajectories exceeds its available modelling capacity κ_L . Formally, if X is the relevant state information and $\ell(X)$ is its encoding complexity, then L is bounded where*

$$\ell(X) > \kappa_L.$$

Boundedness forces compression. If a system could encode the entire regime perfectly and costlessly, it would not need a model in the Mirror sense. It would simply duplicate the relevant state. Real observers do not do this. They preserve selected differences.

Definition 7.3 (Compression operator). *A compression operator for L is a map*

$$\pi_L : S \rightarrow \mathcal{M}_L$$

from regime states to a model space $\mathcal{M}_L$, satisfying $\ell(\pi_L(s)) \leq \kappa_L$, where ℓ measures encoding complexity.

Definition 7.4 (Viability relevance). *A compressed variable $X = \pi_L(s)$ is viability-relevant for L if conditioning on X changes expected future viability:*

$$\mathbb{E}[V_L \mid X] \neq \mathbb{E}[V_L],$$

where V_L is the local viability functional for L .

The central question is not whether a system compresses. Many systems compress. The question is whether the compression preserves distinctions that matter for prediction, correction or continued organisation.

8 Models, self-models and reliability variables

Definition 8.1 (World-model). *A world-model for L is a compressed structure M_E encoding environment-relevant variables such that M_E carries predictive or corrective information about future states or viability of L .*

Definition 8.2 (Self-model). *A self-model for L is a compressed structure M_L encoding internal, operational or system-relative variables of L that are relevant to prediction, correction or viability.*

Definition 8.3 (Reliability variable). *A reliability variable R_L is a variable, estimate or model component tracking the error profile, uncertainty, adequacy or calibration of M_L .*

Examples include confidence calibration, sensor-health estimation, memory-integrity scoring, body-state uncertainty, introspective uncertainty, metacognitive reliability and instrument reliability. In a symbolic agent, R_L may be an explicit integrity score. In a biological system, it may be distributed across metacognitive and interoceptive processes.

Definition 8.4 (Recursive self-modelling). *A subsystem L exhibits recursive self-modelling where R_L is itself represented, updated or used by the system in a way that affects future prediction, correction or viability.*

8.1 Minimal reliability variables

For computation, the reliability variable R_L must not remain merely verbal. The following hierarchy gives toy cases that can be implemented in artificial agents.

Level 0: none.

The system has a world-model and perhaps a self-model, but no variable estimating the reliability of that self-model.

Level 1: scalar confidence.

The system stores a single confidence score for its self-model, such as $r_t \in [0, 1]$.

Level 2: error variance.

The system estimates uncertainty or variance in self-state estimates, for example $R_L = \text{Var}(\hat{s}_L - s_L)$.

Level 3: posterior reliability.

The system maintains a posterior distribution over self-state or model parameters, $p(\theta_L \mid D)$, and uses that posterior in action or correction.

Level 4: meta-controller.

The system contains a controller that decides when to trust, repair, recalibrate or replace the current self-model.

A minimal R_L satisfies the salience condition only if it is not merely recorded but operationally coupled to future action, correction or viability. A scalar confidence score that never changes policy is not enough. A variance estimate that causes the system to slow down, recalibrate sensors, seek information or avoid risky action may be enough in a suitable task. A full posterior is stronger, but not required by definition. The calculus is therefore intentionally implementation-neutral: it asks whether reliability information changes expected viability after cost, not whether the system uses a particular Bayesian, symbolic or neural representation.

Example 8.1 (Minimal toy reliability variable). *An agent estimates its own localisation error as $R_L = \sigma_t^2$. If σ_t^2 exceeds a threshold, the agent switches from fast navigation to calibration mode. If this switch improves long-horizon task completion after accounting for delay and energy cost, then R_L is a candidate reliability variable satisfying positive salience gain in that environment.*

9 Constraint salience

The key quantity is the net value of modelling the reliability of one's own self-model.

Definition 9.1 (Constraint salience gain). *Let M_E be a world-model, M_L a self-model, R_L a reliability variable and $K_L(R_L) \geq 0$ the computational or organisational cost of maintaining and updating R_L . Define the constraint salience gain of R_L for L by*

$$\Sigma_L(R_L) = \mathbb{E}[V_L \mid M_E, M_L, R_L] - \mathbb{E}[V_L \mid M_E, M_L] - K_L(R_L).$$

This expression is intentionally minimal. It does not measure consciousness, intelligence, reportability or moral status. It asks whether adding a reliability model of self-modelling improves expected viability after cost, given that world-modelling and self-modelling are already present.

Remark 9.1. *A more elaborate formulation could replace conditional expectations with mutual information, variational free energy, control value, Bayesian decision value, regret reduction or survival probability. The present definition is chosen because it isolates the Mirror threshold: the point at which self-model reliability becomes operationally valuable.*

A useful information-theoretic variant is

$$\Sigma_L^I(R_L) = \mathbf{l}(R_L; V_L \mid M_E, M_L) - K_L(R_L),$$

where $\mathbf{l}(\cdot; \cdot \mid \cdot)$ denotes conditional mutual information [7]. This version says that R_L matters when it carries additional information about future viability beyond what is already available in the world-model and self-model.

10 Relation to active inference, predictive processing and cybernetics

The salience functional is close to existing ideas, and the differences should be explicit. Mirror Theory does not deny predictive processing, active inference, information theory or cybernetics. It asks a narrower threshold question: when does self-model reliability become a viability-relevant object of correction?

10.1 Predictive processing

Predictive processing treats perception and cognition as structured prediction and error correction. Mirror Theory accepts the importance of prediction, but prediction alone does not license observerhood. A system may minimise prediction error over environmental variables without representing its own self-condition or the reliability of that representation. Mirror adds a recursive condition: prediction becomes observer-relevant where error about the system's own situated modelling condition affects future correction or viability.

10.2 Active inference

Active inference minimises expected free energy and often includes epistemic value, uncertainty reduction and action-perception coupling [10]. The mutual-information version

$$\Sigma_L^I(R_L) = \mathbf{l}(R_L; V_L \mid M_E, M_L) - K_L(R_L)$$

resembles an epistemic value term, but its target is narrower. It is not information-seeking about the world generally. It is information about the reliability of the self-model, conditional on world-modelling and self-modelling already being present. Mirror can therefore be read as identifying a possible subclass or threshold within active-inference architectures: those in which self-model reliability itself becomes action-relevant and viability-relevant.

The deliberate non-mapping is also important. Active inference can describe perception-action regulation without claiming that observerhood has been derived. Mirror Theory uses viability and salience to specify when an architecture crosses from adaptive regulation into

recursive observerhood. Whether this is a genuinely distinct threshold or merely a terminological refinement of active inference is an empirical and mathematical question.

10.3 Cybernetics and requisite variety

Cybernetics emphasises feedback, control and regulation. Ashby’s law of requisite variety says, roughly, that a regulator must have sufficient variety to match the disturbances it must control [6]. Mirror Theory inherits this logic but adds two layers. First, the system must regulate not only external disturbance but also self-model error. Second, the reliability of that self-model must itself become a regulated variable. Feedback is therefore necessary in many cases, but not sufficient for observerhood.

10.4 Information theory

Information theory supplies tools for measuring dependence, compression and uncertainty [7]. Mirror does not identify observerhood with mutual information. A sensor can carry information without being an observer. The information-theoretic salience term is useful only when information about self-model reliability changes expected viability after cost. Observerhood is therefore not information alone; it is information embedded in bounded self-maintaining correction.

11 Observer predicates

Definition 11.1 (Mirror observer predicate). *A subsystem L satisfies the Mirror observer predicate, written $\text{Obs}(L)$, where:*

- (i) L is a bounded local subsystem of a constraint regime;
- (ii) L maintains a world-model M_E ;
- (iii) L maintains a self-model M_L ;
- (iv) L maintains or uses a reliability variable R_L for M_L ;
- (v) $\Sigma_L(R_L) > 0$.

This is not a consciousness predicate. It is a structural observerhood predicate. It says that the system’s own self-location or self-condition has become viability-relevant in a way that makes reliability tracking worth its cost.

Remark 11.1. *The predicate is intentionally conservative. Systems can be complex, predictive, adaptive or self-referential without satisfying it. The predicate is licensed only when reliability about self-modelling is operationally coupled to future viability.*

12 First results

Theorem 12.1 (Necessary Constraint Theorem). *Let L be a local subsystem. If L satisfies the Mirror observer predicate, then there must exist a constraint regime $C = (S, T, \mathcal{I}, V, \kappa)$ such that:*

- (i) L is embedded in, or represented by, a bounded local part of S ;

- (ii) *admissible transitions preserve enough invariant structure for persistence of L over a non-trivial horizon;*
- (iii) *L compresses relevant regime information under a capacity bound;*
- (iv) *the reliability of L 's self-model is viability-relevant after cost.*

Proof. By definition, $\text{Obs}(L)$ requires a bounded local subsystem, a world-model, a self-model, a reliability variable and positive salience gain. A bounded local subsystem requires a state space and transition structure in which locality and persistence can be stated. A world-model and self-model require compression under a capacity bound. Positive salience gain requires a viability functional and a cost term. If admissible transitions preserve no relevant invariant structure, there is no persistent organisation to which M_L can refer. If there is no capacity bound, compression is not forced. If reliability information is not viability-relevant after cost, $\Sigma_L(R_L) \leq 0$ and the observer predicate fails. Therefore all listed conditions are necessary. $\square$

Corollary 12.1 (Exclusion of trivial systems). *A rock, thermostat, self-referential sentence, ordinary feedback loop or predictor does not satisfy the Mirror observer predicate merely by persisting, responding, referring to itself or predicting. It must satisfy the full salience condition $\Sigma_L(R_L) > 0$.*

Theorem 12.2 (Constraint-Salience Threshold). *Let L be a bounded local subsystem with world-model M_E , self-model M_L , reliability variable R_L , viability functional V_L and reliability cost $K_L(R_L)$. If*

$$\Sigma_L(R_L) > 0,$$

then error about L 's own self-condition, self-location or self-model reliability is viability-relevant after cost. In that case L crosses the Mirror threshold from ordinary self-modelling into candidate recursive observerhood. If $\Sigma_L(R_L) \leq 0$, then L may remain a predictor or self-modelling agent, but the recursive observer predicate is not licensed.

Proof. Assume $\Sigma_L(R_L) > 0$. By definition,

$$\mathbb{E}[V_L \mid M_E, M_L, R_L] - \mathbb{E}[V_L \mid M_E, M_L] > K_L(R_L) \geq 0.$$

Thus adding R_L changes expected viability by more than its cost. Since R_L tracks the adequacy or error profile of the self-model, the reliability of self-modelling is not merely externally describable; it is operationally valuable for the system. Error about the system's own condition, location or modelling adequacy becomes part of the correction problem. This is precisely the Mirror threshold.

Conversely, if $\Sigma_L(R_L) \leq 0$, the addition of R_L fails to improve expected viability after cost. The system may still model the world and itself, but recursive reliability tracking is not selected by the salience criterion. The observer predicate therefore remains unlicensed. $\square$

Corollary 12.2 (Perturbation prediction). *Where $\Sigma_L(R_L) > 0$, targeted perturbation or ablation of R_L should produce differential degradation in correction, persistence or long-horizon performance under conditions where self-error matters. Where no such degradation is observed under appropriate conditions, the classification of L as a Mirror observer is weakened.*

13 Admissible transformations and observer equivalence

Observer identity is not material sameness. It is preservation of the relevant observer-maintaining constraints under transformation.

Definition 13.1 (Admissible transformation). *Let $C = (S, T, \mathcal{I}, V, \kappa)$ and $C' = (S', T', \mathcal{I}', V', \kappa')$. A map $F : S \rightarrow S'$ is an admissible transformation from C to C' if it preserves transition structure, relevant invariants and viability ordering up to specified equivalence:*

- (i) *if $(s, s') \in T$, then $(F(s), F(s')) \in T'$ or is transition-equivalent;*
- (ii) *for relevant $J \in \mathcal{I}$, there exists $J' \in \mathcal{I}'$ such that $J(s) \sim J'(F(s))$;*
- (iii) *for trajectories τ_1, τ_2 , if $V(\tau_1) \geq V(\tau_2)$, then $V'(F\tau_1) \gtrsim V'(F\tau_2)$.*

Definition 13.2 (Observer equivalence). *Two subsystems $L \subset C$ and $L' \subset C'$ are observer-equivalent, written $L \simeq_{obs} L'$, if there exists an admissible transformation mapping (M_E, M_L, R_L) to (M'_E, M'_L, R'_L) while preserving positive salience gain and the relevant observer-maintaining invariants.*

Proposition 13.1 (Substrate independence under admissible transformation). *If $L \simeq_{obs} L'$, then L and L' instantiate the same observer-type at the level of Mirror structure, even if their substrates differ.*

Proof. Observer equivalence requires preservation of the world-model, self-model, reliability structure and positive salience gain under an admissible transformation. Since the Mirror observer predicate depends on these structural conditions rather than on substrate identity, both systems instantiate the same observer-type relative to the specified equivalence. $\square$

14 Regime collapse and boundary cases

Definition 14.1 (Regime collapse). *A constraint regime C undergoes collapse relative to L when the invariants and transition structures required for L 's observer-maintaining organisation are no longer preserved along admissible trajectories.*

Definition 14.2 (Local observer death). *A local observer L undergoes observer death when $\text{Obs}(L)$ irreversibly fails because the organisation required to maintain (M_E, M_L, R_L) and positive salience gain collapses.*

This formalises a point from the ontology paper. The matter or substrate may persist, but the observer-maintaining constraint structure may fail. The system's components can continue in transformed form without preserving observerhood.

Boundary cases are important because they show that the Necessary Constraint Theorem is not simply a tautology. Many systems satisfy some premises while failing the full observer predicate. A non-trivial failure case is especially useful: an advanced predictor may have a rich environmental model and high task performance while failing the observer predicate because errors about its own modelling condition never become variables in its own correction. Conversely, a less capable agent may satisfy a weak Mirror predicate if its own sensor reliability, memory corruption or bodily state directly affects long-horizon viability and is used in policy selection.

15 Diagram of the calculus

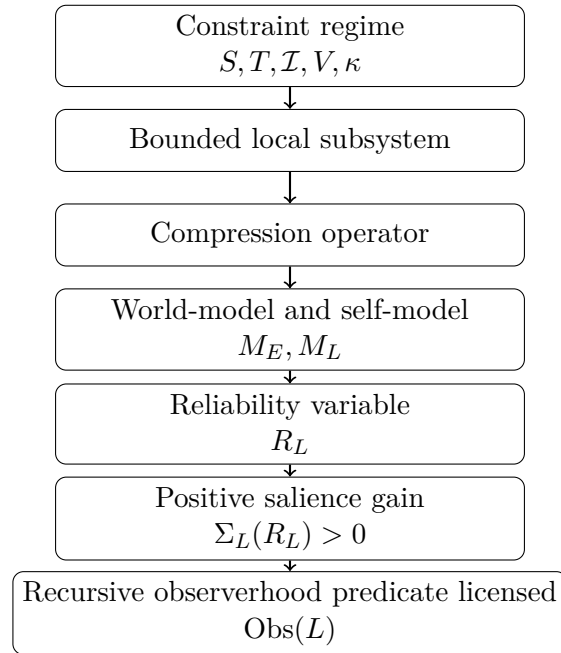


Figure 1: Constraint calculus dependency structure. The diagram is not a developmental law. It states the formal dependencies required before the Mirror observer predicate is licensed.

System	World-model	Self-model	Reliability variable	Mirror status
Thermostat	Minimal	No	No	Feedback without observerhood
PID controller	Limited	No	No	Regulation without self-location
Self-referential sentence	No	Syntactic only	No	Self-reference without viability
Ordinary predictor	Yes	No or incidental	No	Prediction without recursive self-modelling
Self-describing program	Partial	Code representation	Usually no	Self-description without viability-relevant reliability
LLM chatbot	Broad linguistic model	Partial / unstable	Usually external	Borderline unless persistent self-model and reliability coupling is added
Mirror-style agent	Symbolic world-field	Explicit self-model	Integrity / reliability score	Candidate if reliability affects correction under task conditions
Human observer	Yes	Yes	Metacognition, interoception, confidence	Paradigmatic biological case

Table 2: Boundary systems. The table separates feedback, prediction, self-reference and self-description from recursive observerhood.

16 Toward future physics

The calculus does not yet modify physics, but it indicates what a future physics of constraint would have to do.

First, it would have to identify physical analogues of $S, T, \mathcal{I}, V, \kappa$. In ordinary physics, state spaces and transition rules are already central. The new burden is to explain whether viability-like persistence functionals and capacity bounds can be defined without importing observers by hand.

Second, it would have to connect invariance and modelling. Relativity, for example, is built around invariant structure. Quantum theory is built around state spaces, transformations, observables and probabilities. A constraint-theoretic extension would have to show that Mirror variables are not merely metaphors placed on top of these theories.

Third, it would have to produce a new prediction. A candidate relation such as

$$\Phi_L = \mathbb{I}(R_L; V_L \mid M_E, M_L) - K_L(R_L)$$

may be meaningful for agents, but it is not yet a spacetime field. Before any equation can couple Φ_L to causal geometry, one must define what Φ_L measures physically, how it transforms, what units it has, and why existing experiments have not already ruled it out.

Remark 16.1 (Discipline condition). *A Mirror-compatible physics must satisfy five constraints: it must reduce to tested physics in known domains; define its new variables operationally; show how any proposed constraint-salience quantity transforms under the relevant lower-level symmetries, such as Lorentz or gauge symmetries, or else derive it as an emergent coarse-grained quantity; derive at least one prediction not already implied by existing physics; and specify an experiment or observation that could falsify the prediction.*

The symmetry requirement is essential. If a future quantity Φ_L is claimed to correspond to physical constraint salience, then one of two things must be shown. Either Φ_L is a lower-level physical quantity with well-defined transformation behaviour under the symmetries of the theory in question, or Φ_L is an emergent variable defined only after coarse-graining, like temperature or entropy. Without this step, a proposed physical extension would merely attach new language to established physics without earning physical status.

17 Relationship to the Mirror Programme

This paper provides the formal entry point for the Mathematics arc of Volume I. It inherits the programme-level spine from MP-00 and the observerhood target from the Volume I overview [1, 2]. Its role is to turn the Theory arc’s conceptual derivation into a set of formal objects: constraint regimes, local subsystems, compression operators, viability functionals, self-models, reliability variables and salience thresholds.

The paper also establishes the bridge to the remainder of Volume I. The next mathematics paper can sharpen the minimal observer threshold by specifying when reliability is represented, action-guiding, cost-sensitive and positively coupled to viability. The memory-governance paper can then ask how an observer that satisfies the predicate at one time remains the same observer across perturbation and update. The Observerhood Labs can operationalise $\Sigma_L(R_L)$ through ablation, perturbation and repair tasks. The Physics support arc can ask what physical conditions are required for causal update, memory preservation, thermodynamic maintenance and temporal asymmetry.

The unifying rule is the same throughout: do not treat observerhood as a primitive. Derive the conditions under which it becomes a maintained, repairable and viability-relevant organisation.

18 Limitations

The limitations are substantial.

First, the calculus is architectural. It supplies definitions and threshold conditions, but it does not by itself prove that any particular biological, artificial or physical system satisfies them.

Second, the viability functional must be specified domain by domain. Without a measurable V_L , salience gain remains formal notation.

Third, positive salience gain is not a complete theory of mind. It is a structural observerhood threshold. It does not settle phenomenal consciousness, moral status, personhood or legal agency.

Fourth, the calculus is compatible with several existing frameworks. It may ultimately be absorbed by a more mature active-inference, control-theoretic or information-theoretic account. The value of Mirror Mathematics depends on whether its threshold structure clarifies cases that those frameworks leave ambiguous.

Fifth, the physical implications are only support conditions. This paper does not assert new physics, does not derive the value of c , does not modify thermodynamics and does not identify constraint salience with a physical field.

19 Research programme

The immediate research programme is mathematical and computational rather than physical.

19.1 Mathematical tasks

1. Formalise classes of constraint regimes: deterministic, stochastic, continuous, discrete, symbolic and hybrid.
2. Define invariant preservation under approximate equivalence rather than exact equality.
3. Develop observer equivalence classes and morphisms between regimes.
4. Study fixed points and attractors of salience gain under repeated self-model correction.
5. Relate Σ_L to conditional mutual information, regret reduction, expected utility, free energy and control value.

19.2 Computational tasks

1. Build three artificial agent classes in the same environment: world-model only; world-model plus self-model; and world-model plus self-model plus reliability tracking.
2. Implement minimal reliability variables at increasing levels: scalar confidence, variance estimate, posterior uncertainty and meta-controller.
3. Introduce sensor degradation, memory corruption, false self-location, actuator impairment and misleading internal confidence.

4. Perturb or ablate R_L under tasks where self-error matters and under tasks where self-error does not matter.
5. Measure ablation loss, recovery time, reliability calibration, long-horizon correction, task persistence and counterfactual coupling between R_L and action.
6. Estimate $\Sigma_L(R_L)$ or an information-theoretic proxy in each agent class.
7. Use exclusion tests to distinguish recursive observerhood from feedback, ordinary prediction, self-reference and self-description.

Progress would consist in finding a task family where reliability-tracking agents outperform self-model-only agents specifically when self-error is viability-relevant, and where ablation of R_L produces differential degradation. A negative result would be equally informative: if R_L adds no measurable value under conditions where the theory says it should, the salience threshold requires revision.

19.3 Physical tasks

Physical work should come later. It would require:

1. identifying physical quantities corresponding to constraint salience;
2. showing transformation laws for those quantities;
3. proving compatibility with relativity and quantum theory in existing domains;
4. deriving one new testable prediction;
5. specifying a falsification pathway.

20 Conclusion

Mirror Mathematics I provides a formal grammar for the Mirror Programme. It does not discover a new force, alter relativity, explain consciousness or derive the origin of reality. Its contribution is more modest and more necessary: it defines what must be meant by constraint, persistence, compression, self-modelling, reliability and salience before those words can responsibly enter either artificial-agent experiments or speculative physics.

The central object is the constraint regime $C = (S, T, \mathcal{I}, V, \kappa)$. The central operational burden is specifying V_L in measurable terms. The central threshold is $\Sigma_L(R_L) > 0$. The central exclusion is that persistence, feedback, prediction and self-reference are not enough. Recursive observerhood is licensed only where bounded systems model the world, model themselves, and gain viability by modelling the reliability of that self-location.

If future work remains philosophical, this calculus offers cleaner definitions. If future work becomes computational, it offers metrics and experiments. If future work ever becomes physical, it supplies the precondition: do not write new physics until the constraint variables are defined, transformation laws are known, and falsifiable predictions can be stated.

In that sense, the paper marks a pause rather than an escalation. Mirror Theory has reached the edge of physics. The next responsible step is not to leap over that edge, but to build the mathematical bridge.

A Glossary

Constraint regime

A state space equipped with transition structure, invariants, viability functional and capacity bound.

Invariant

A function, relation or structure preserved under admissible transformations.

Viability

Organisational preservation relative to a system-type, not moral value.

Compression

Selective preservation of distinctions under bounded capacity.

World-model

A compressed model of environment-relevant variables.

Self-model

A compressed model of internal or system-relative variables.

Reliability variable

A variable tracking the adequacy, uncertainty or error profile of the self-model.

Salience gain

The net viability benefit of modelling self-model reliability after cost.

Observer predicate

The structural condition under which recursive observerhood is licensed.

Observer equivalence

Preservation of observer-maintaining constraints across admissible transformations.

B Minimal notation table

Symbol	Meaning
S	State space
T	Admissible transition relation or operator
$\mathcal{I}$	Family of invariants
V	Viability or persistence functional over a horizon
κ	Capacity bound
L	Local subsystem
M_E	World-model
M_L	Self-model
R_L	Reliability variable for self-model
$K_L(R_L)$	Cost of maintaining reliability variable
$\Sigma_L(R_L)$	Constraint salience gain
$\text{Obs}(L)$	Mirror observer predicate

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